Sensitivity analysis and density estimation using the Malliavin calculus

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Consider $(F^\zeta)_\zeta$ a family of random variables depending on a parameter ζ.

$$\frac{\partial}{\partial\zeta}\mathbf{E}\left[f\left(F^\zeta\right)\right] = \begin{cases} \mathbf{E}\left[f'\left(F^\zeta\right)\frac{\partial}{\partial\zeta}F^\zeta\right] \\ \simeq \frac{\mathbf{E}\left[f\left(F^{\zeta+h}\right)\right] - \mathbf{E}\left[f\left(F^{\zeta-h}\right)\right]}{2h} \end{cases}$$

Expectations can be computed by the Monte Carlo method:

$$E[F] \simeq \frac{F_1 + \cdots + F_n}{n},$$

where $F_1, \ldots, F_n$ is a random sample of F.

Application (1): sensitivity analysis in finance - Greeks

Price process:

$$\frac{dS_t^\zeta}{S_t^\zeta} = r(S_t^\zeta)dt + \sigma(S_t^\zeta)dM_t, \qquad S_0^\zeta = x.$$

$F^\zeta = S_T$ and $\zeta \in \{x, r, \sigma, T, ...\}$.

Payoff function:

- $f(x) = (x - K)^+$

- $f(x) = 1_{[K,\infty)}(x)$.

Option price:

$E[f(S_T^\zeta)]$.

Greeks:

- $\zeta = x$: Delta
- $\zeta = \sigma$: Vega
- $\zeta = r$: Rho
- $\zeta = T$: Theta.

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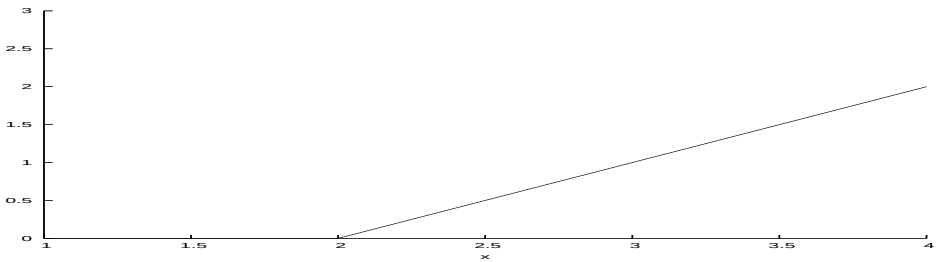
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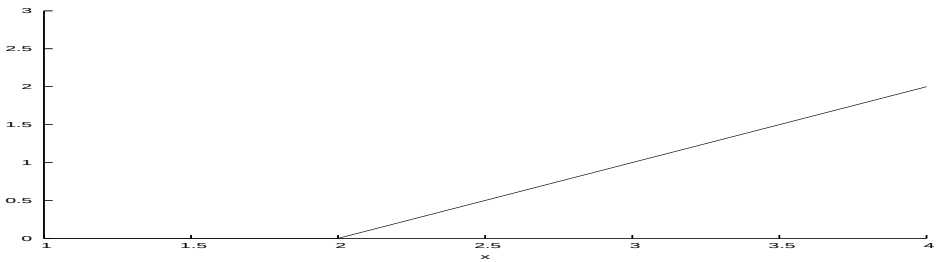
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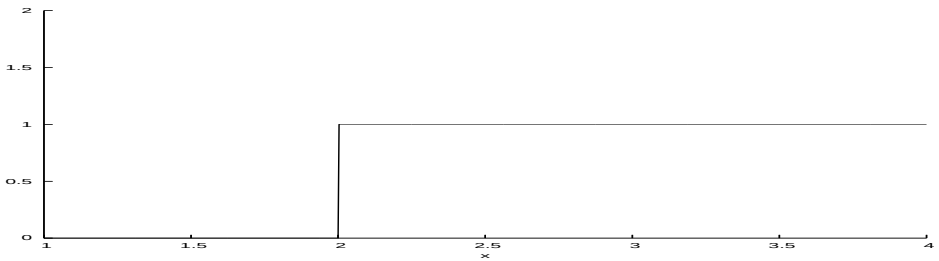
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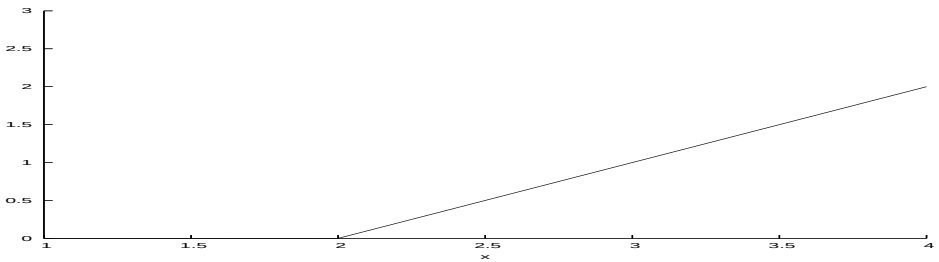
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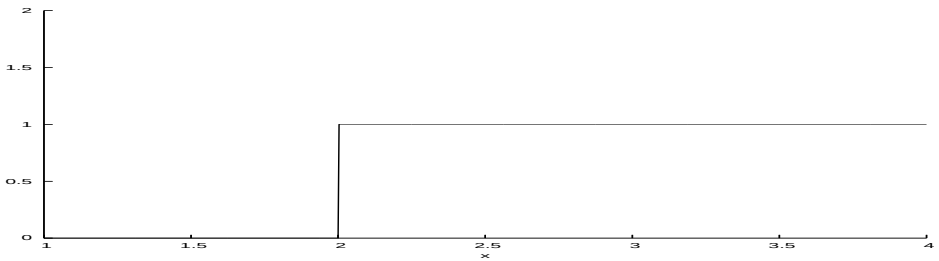
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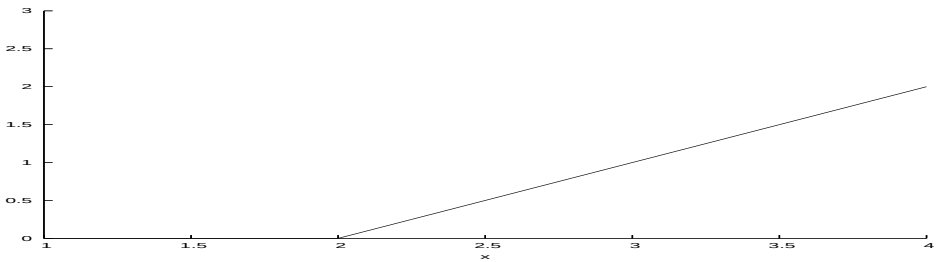
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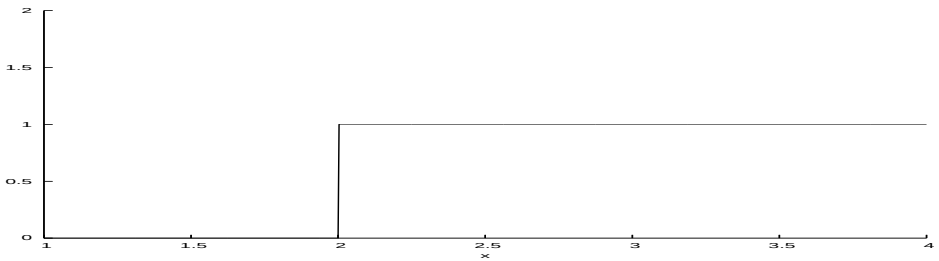
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Application (2): density estimation

Let $F^\zeta = F - \zeta$ and $f(x) = 1_{(-\infty,0]}$.

Density of F:

$$\begin{aligned}\phi_F(\xi) : &= \frac{\partial}{\partial\zeta} P(F \le \zeta) = \frac{\partial}{\partial\zeta} \mathbf{E}\left[1_{(-\infty,0]}(F-\zeta)\right] = \frac{\partial}{\partial\zeta} \mathbf{E}\left[f\left(F^\zeta\right)\right] \\ &\simeq \frac{\mathbf{E}\left[f\left(F-(\zeta+h)\right)\right] - \mathbf{E}\left[f\left(F-(\zeta-h)\right)\right]}{2h} = \frac{\mathbf{E}\left[1_{[\zeta-h,\zeta+h]}(F)\right]}{2h}.\end{aligned}$$

Example: $F = \int_0^T e^{-rt} dN_t$.

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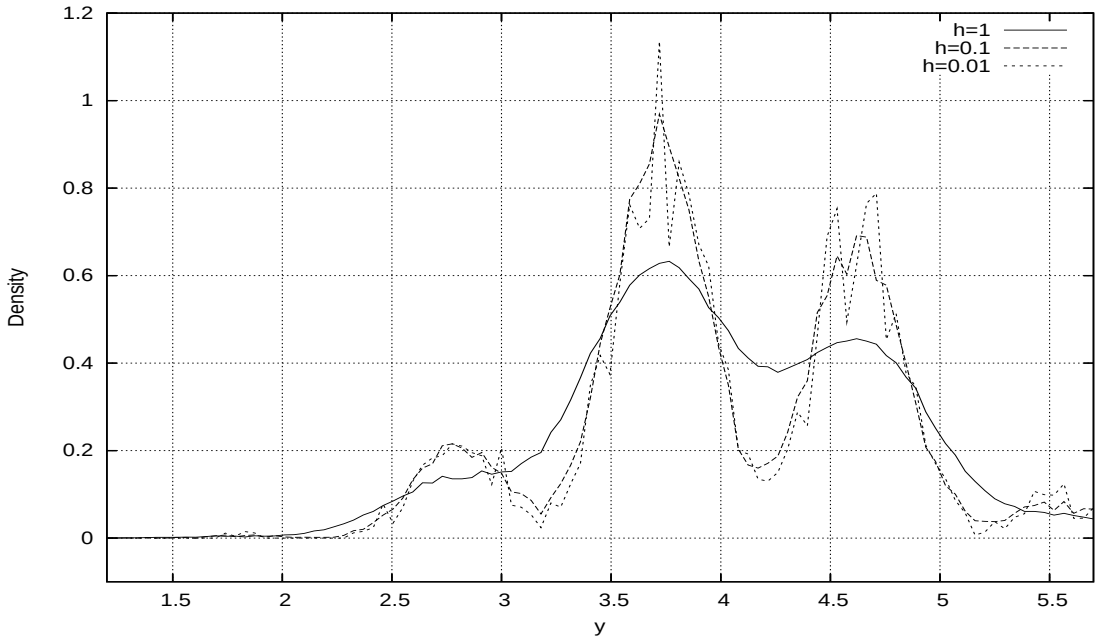
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$$f'(F^\zeta) = \frac{D_w\left[f(F^\zeta)\right]}{D_w F^\zeta}.$$

2. D_w^* the adjoint of D_w:

$$\langle F, D_w G\rangle = E[FD_w G] = E[GD_w^* F] = \langle G, D_w^* F\rangle.$$

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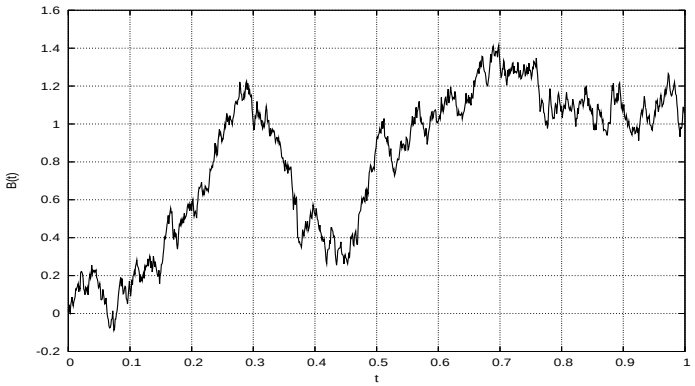
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Wiener case [FLL+99], [FLLL01], [Ben02]

- $M_t = B_t$ is a Brownian motion:



- $\frac{dS_t^\zeta}{S_t^\zeta} = \sigma(S_t^\zeta)dB_t + r(S_t^\zeta)dt,\ S_0^\zeta = x.$
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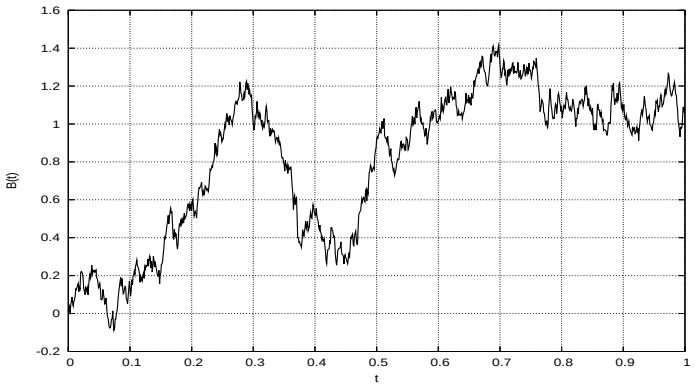
$$D_w f(B_{t_1}, \ldots, B_{t_n}) = \sum_{i=1}^{n} \int_0^{t_i} w_s ds \frac{\partial f}{\partial x_i}(B_{t_1}, \ldots, B_{t_n}).$$

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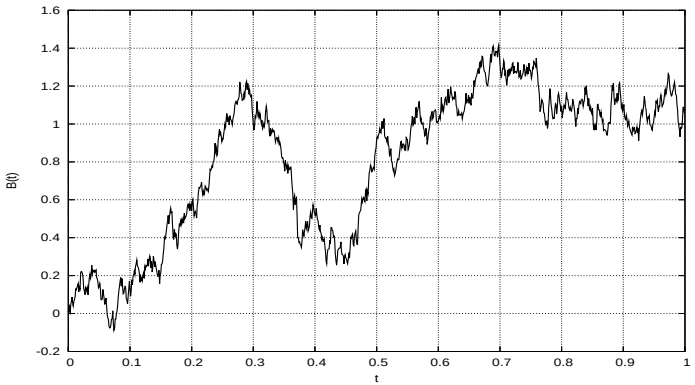
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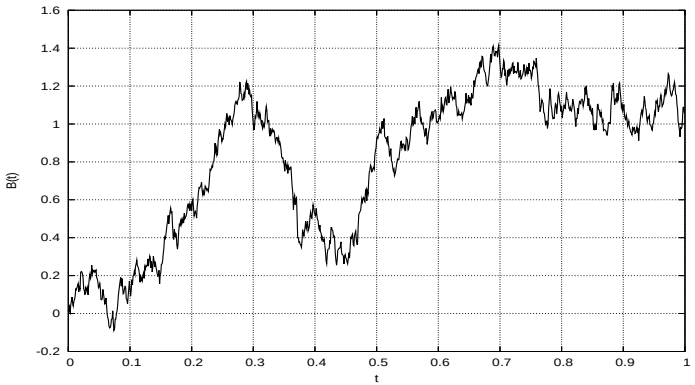
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In [FLL+99] the relations

$$f'(F^\zeta) = \frac{D_t\left[f(F^\zeta)\right]}{D_t F^\zeta}, \qquad 0 \leq t \leq T, \ a.s.,$$

and

$$D_t S_T^x = \frac{\partial_x S_T^x}{\partial_x S_t^x}\sigma(S_t^x), \qquad 0 \leq t \leq T, \ a.s.,$$

are used, cf. [Nua95]. This gives

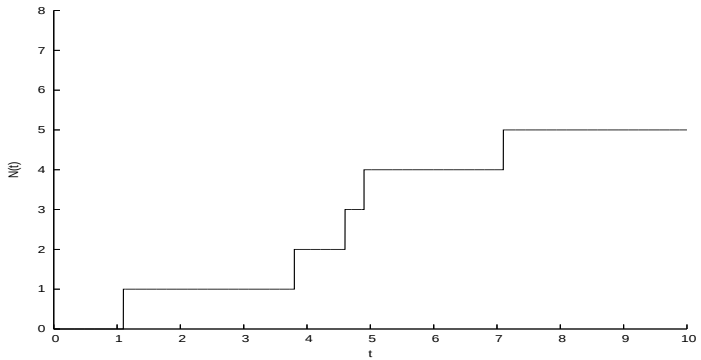
$$\partial_x S_T^x f'(S_T^x) = \frac{\partial_x S_t^x}{\sigma(S_t^x)} D_t f(S_T^x), \qquad 0 \leq t \leq T, \ a.s., \tag{2}$$

which implies if $\int_0^T w_s ds = 1$:

$$\begin{aligned}
\text{Delta} &= \frac{\partial}{\partial x}\mathbf{E}\left[f\left(S_T^x\right)\right] = \mathbf{E}\left[\partial_x S_T^x f'\left(S_T^x\right)\right] = \mathbb{E}\left[\int_0^T w_t \frac{\partial_x S_t^x}{\sigma(S_t^x)} D_t f(S_T^x) dt\right] \\
&= \mathbb{E}\left[f(S_T^x)\delta\left(1_{[0,T]} w \frac{\partial_x S^x}{\sigma(S^x)}\right)\right] = \mathbb{E}\left[f(S_T^x)\int_0^T w_t \frac{\partial_x S_t^x}{\sigma(S_t^x)} dB_t\right] \\
&= \mathbf{E}\left[f\left(S_T^x\right)\frac{B_T}{\sigma x T}\right].
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Poisson case

- $M_t = N_t$ is a Poisson process with jump times $T_1, T_2, T_3, \ldots,$



- $w \in \mathcal{C}_c^1((0,\infty))$, and

$$D_w F = -\sum_{n=1}^{\infty} 1_{\{N_T=n\}} \sum_{i=1}^{n} \frac{\partial f_n}{\partial x_i}(T_1, \ldots, T_n),$$

for F of the form

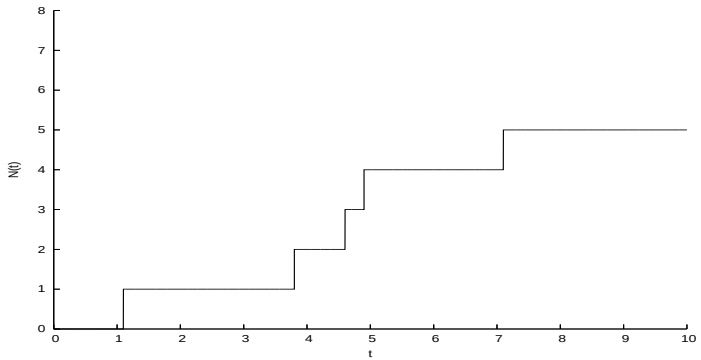
$$F = f_0 1_{\{N_T=0\}} + \sum_{n=1}^{\infty} 1_{\{N_T=n\}} f_n(T_1, \ldots, T_n).$$

- Recall that

$$E[F] = f_0 e^{-\lambda T} + e^{-\lambda T} \sum_{n=1}^{\infty} \frac{1}{n!} \int_0^T \cdots \int_0^T f_n(t_1, \ldots, t_n) dt_1 \cdots dt_n.$$

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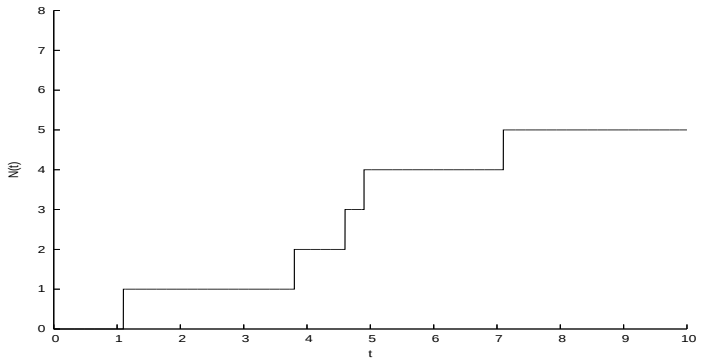
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$$F = f_0 1_{\{N_T=0\}} + \sum_{n=1}^{\infty} 1_{\{N_T=n\}} f_n(T_1, \ldots, T_n).$$

- Recall that

$$E[F] = f_0 e^{-\lambda T} + e^{-\lambda T} \sum_{n=1}^{\infty} \frac{1}{n!} \int_0^T \cdots \int_0^T f_n(t_1, \ldots, t_n) dt_1 \cdots dt_n.$$

By standard integration by parts we have, under the boundary condition $w(0) = w(T) = 0$:

$$\begin{aligned} E[D_w F] &= -e^{-\lambda T} \sum_{n=1}^{m} \frac{\lambda^n}{n!} \int_0^T \cdots \int_0^T \sum_{k=1}^{n} w(t_k) \partial_k f_n(t_1, \ldots, t_n) dt_1 \cdots dt_n \\ &= e^{-\lambda T} \sum_{n=1}^{m} \frac{\lambda^n}{n!} \int_0^T \cdots \int_0^T f_n(t_1, \ldots, t_n) \sum_{k=1}^{n} \dot{w}(t_k) dt_1 \cdots dt_n \\ &= E\left[F \sum_{k=1}^{k=N(T)} \dot{w}(T_k) \right] = E\left[F \int_0^T \dot{w}(t) dN(t) \right]. \end{aligned}$$

Next, letting

$$D_w^* G = G \int_0^T \dot{w}(t) dN(t) - D_w G$$

we get

$$\begin{aligned} E[GD_w F] = E[D_w(FG) - FD_w G] &= E\left[F \left(G \int_0^T \dot{w}(t) dN(t) - D_w G \right) \right] \\ &= E[FD_w^* G]. \end{aligned}$$

Asian options and reserve processes [KP04], [PW04]

- Price process: $S_t^\zeta = S_0^\zeta e^{rt}(1+\sigma)^{N_t}$.
- Asian options: $F^\zeta = \frac{1}{T}\int_0^T S_t^\zeta dt$.

$$W_\Delta = \frac{-1}{x\sigma}\left(1 - \frac{\int_0^T S_t^\zeta dt \int_0^T \dot{w}_t d\tilde{N}_t}{\int_0^T w_t S_{t-}^\zeta dN_t} + \frac{\int_0^T S_t^\zeta dt \int_0^T w_t\left(\dot{w}_t + r w_t\right) S_{t-}^\zeta dN_t}{\left(\int_0^T w_t S_{t-}^\zeta dN_t\right)^2}\right).$$

- Density of reserve processes: $F = \int_0^T e^{(T-t)r} dN_t$.

$$W_y = \frac{\int_0^T \dot{w}(t) dN(t) + \frac{\int_0^T e^{-rt} w(t)(rw(t) - \dot{w}(t)) dN(t)}{\int_0^T w(t) e^{-rt} dN(t)}}{r\int_0^T w(t) e^{r(T-t)} dX(t)}.$$

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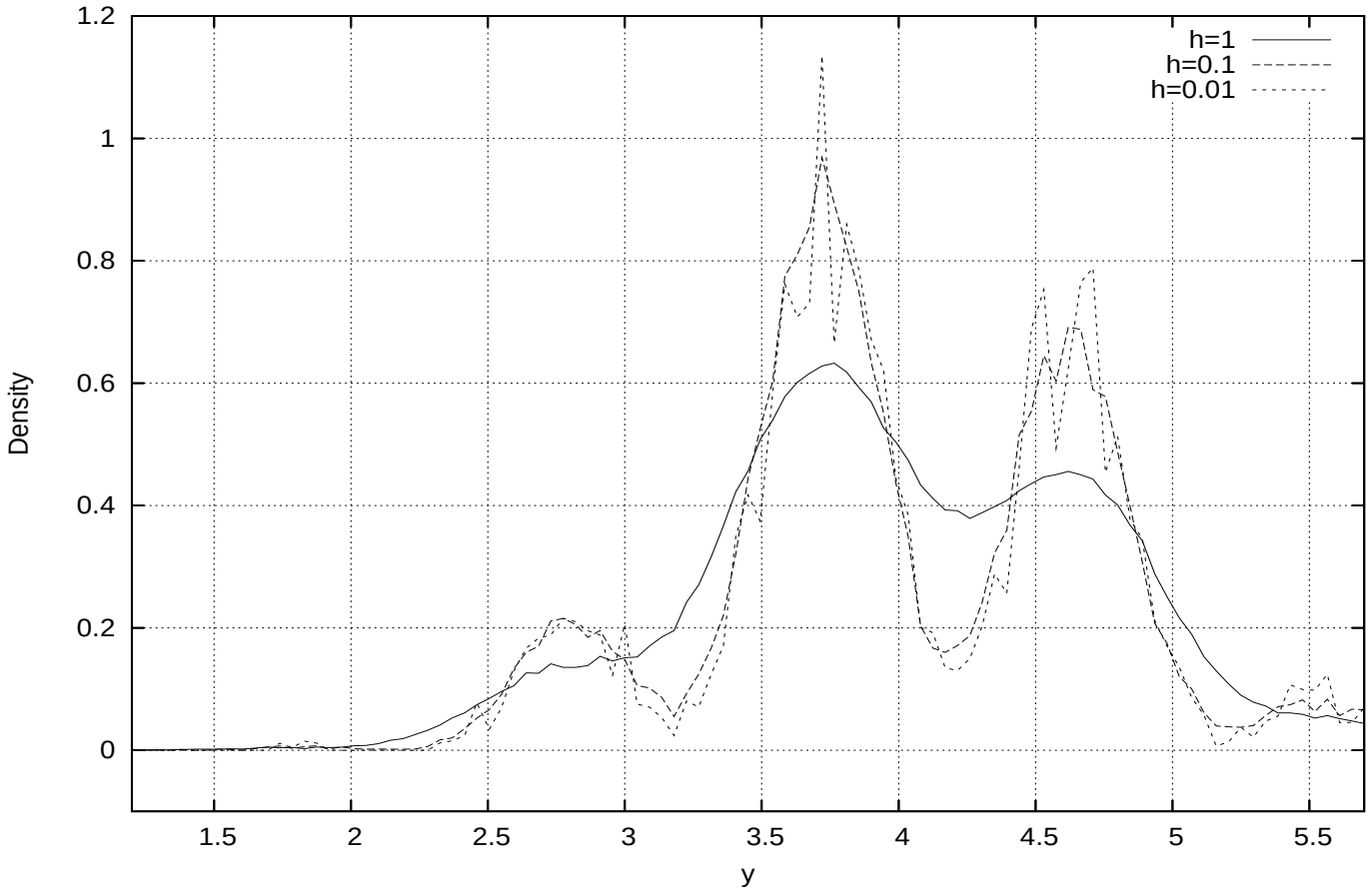
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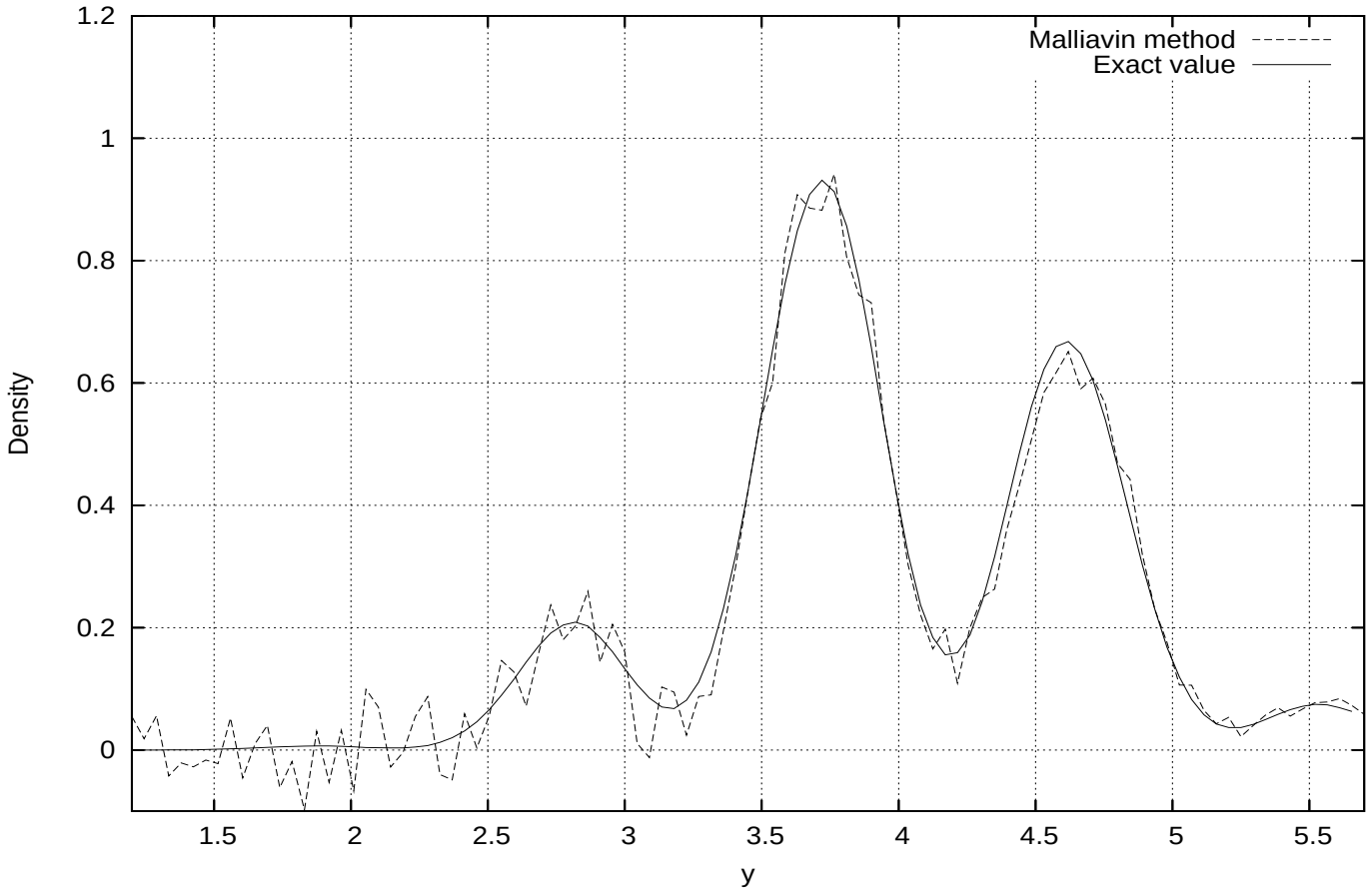
Density estimation - finite differences

$$\phi_F(y) = \frac{\partial}{\partial y}\mathbf{E}\left[1_{(-\infty,0]}(F-y)\right] \simeq \frac{\mathbf{E}\left[1_{[y-h,y+h]}(F)\right]}{2h}.$$



Density estimation - Malliavin method

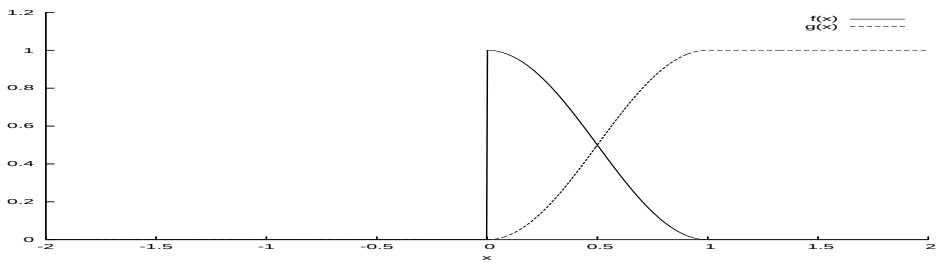
$$\phi_F(y) = \frac{\partial}{\partial y}\mathbf{E}\left[1_{(-\infty,0]}(F-y)\right] = -\mathbf{E}\left[1_{(-\infty,0]}(F-y)D_w^*\left(\frac{1}{D_w F}\right)\right].$$



Consider the decomposition

$$1_{[0,\infty)} = f + g,$$

where g is $\mathcal{C}^1$:

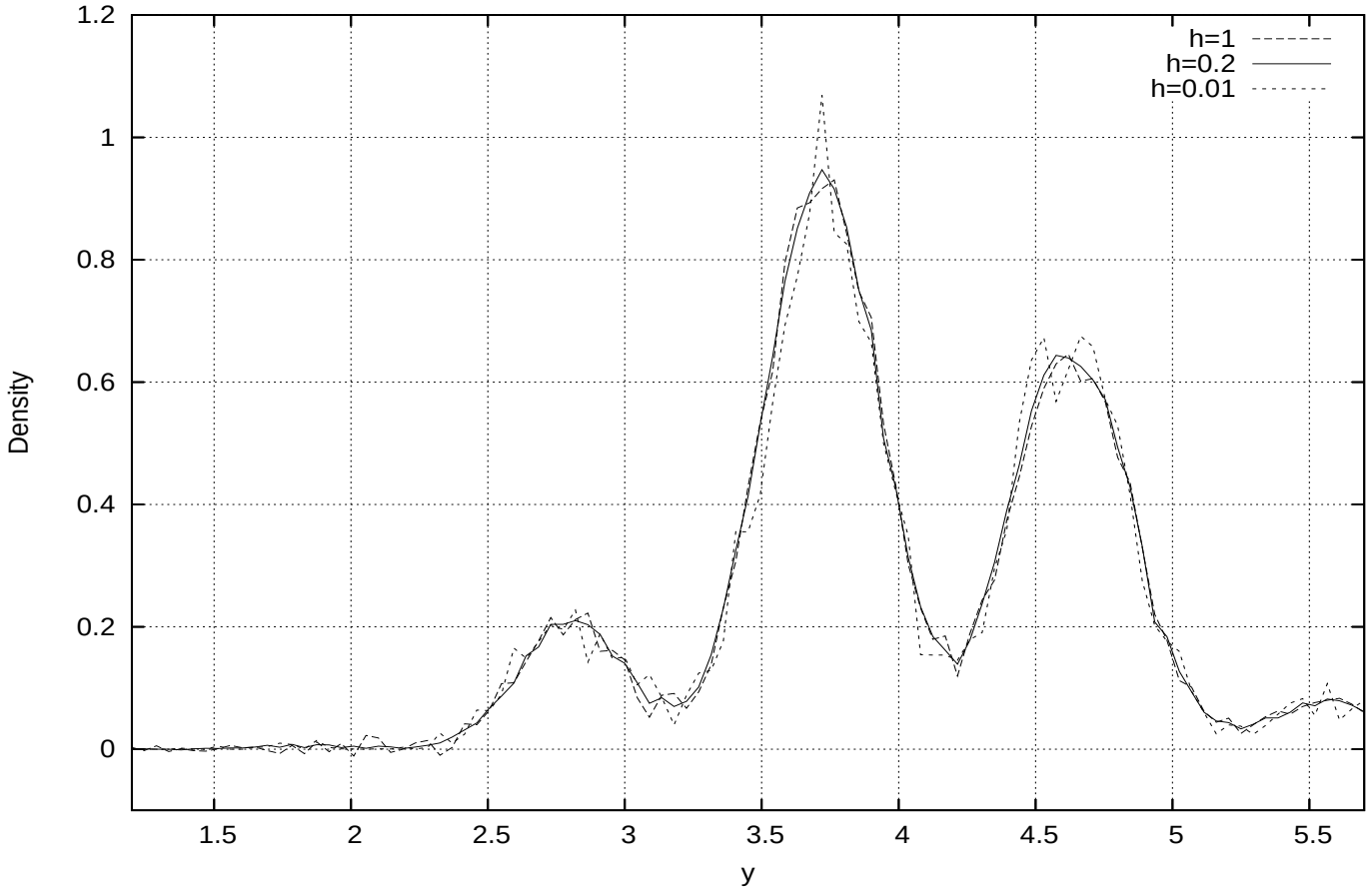


We have

$$\begin{aligned}
\frac{d}{dy}E[1_{[0,\infty)}(F-y)] &= \frac{d}{dy}E\left[f\left(\frac{F-y}{h}\right)\right] + \frac{d}{dy}E\left[g\left(\frac{F-y}{h}\right)\right] \\
&= E\left[D_w^*\left(\frac{1}{D_w F}\right) f\left(\frac{F-y}{h}\right)\right] + \frac{1}{h}E\left[1_{\{F>y\}} f'\left(\frac{F-y}{h}\right)\right].
\end{aligned}$$

Density estimation - localized Malliavin method

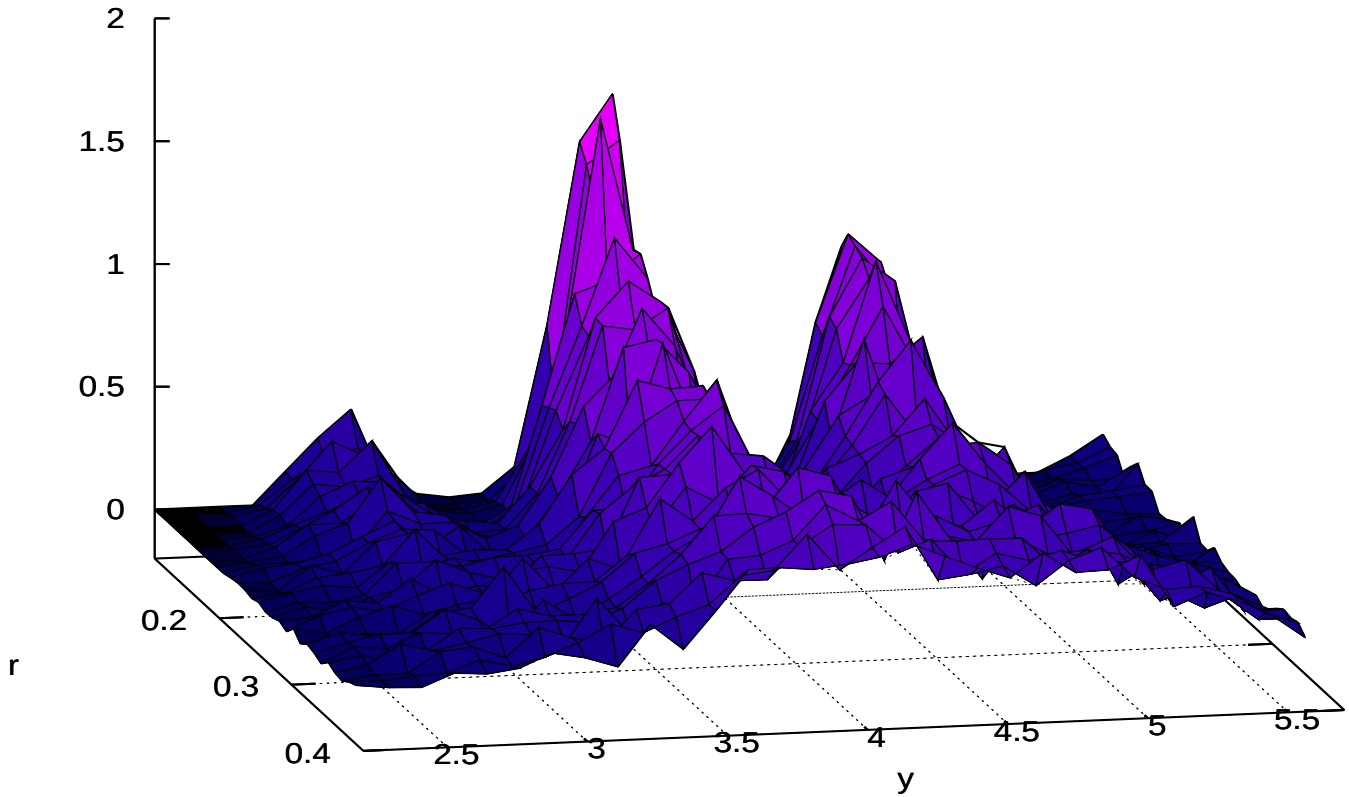
$$\phi_F(y) = -E\left[f\left(\frac{F-y}{h}\right) D_w^*\left(\frac{1}{D_w F}\right)\right] - \frac{1}{h} E\left[1_{\{F>y\}} f'\left(\frac{F-y}{h}\right)\right].$$



Optimization: $f(x) = e^{-x}$, $x \geq 0$, $h = \|W\|^{-1}_{L^2(\Omega)}$.

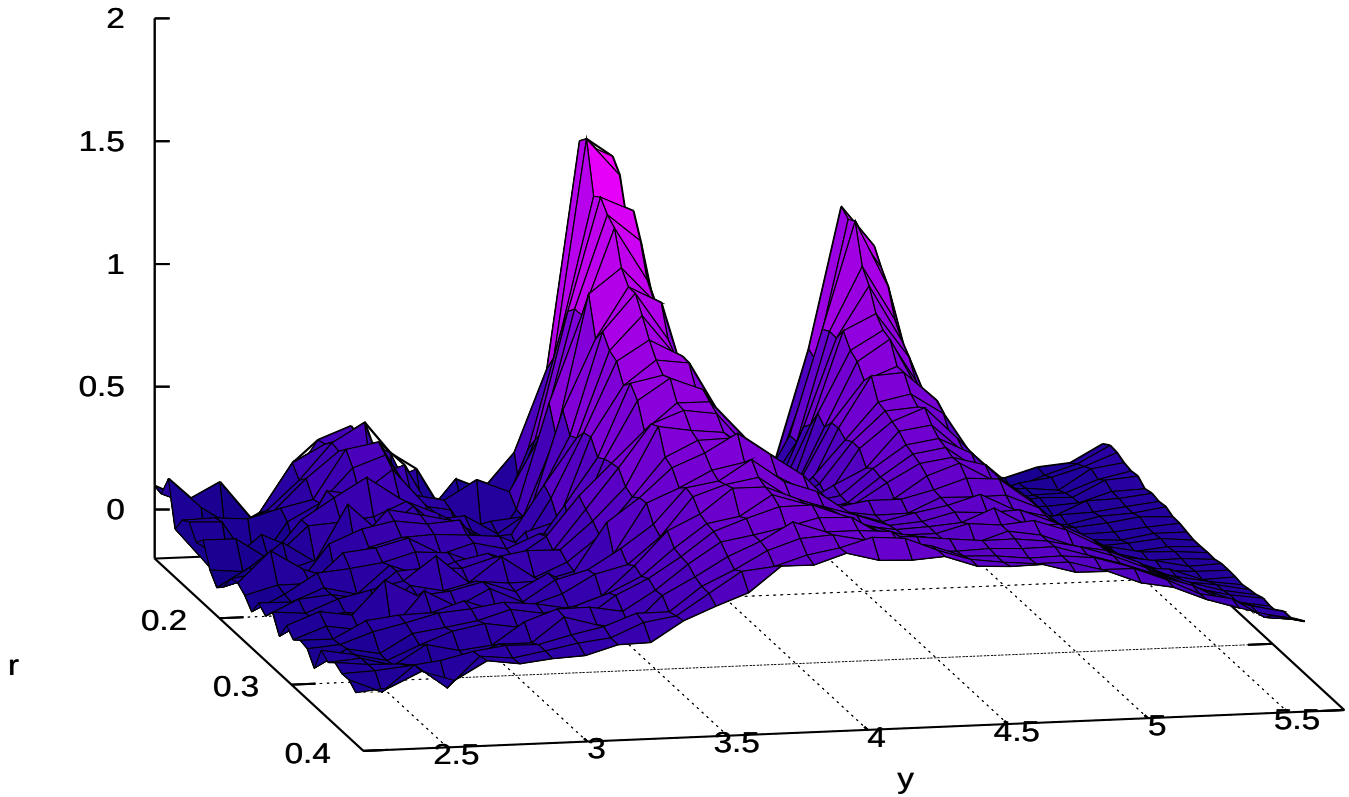
Density estimation - finite differences

$$F = \int_0^T e^{-rt} dN_t.$$



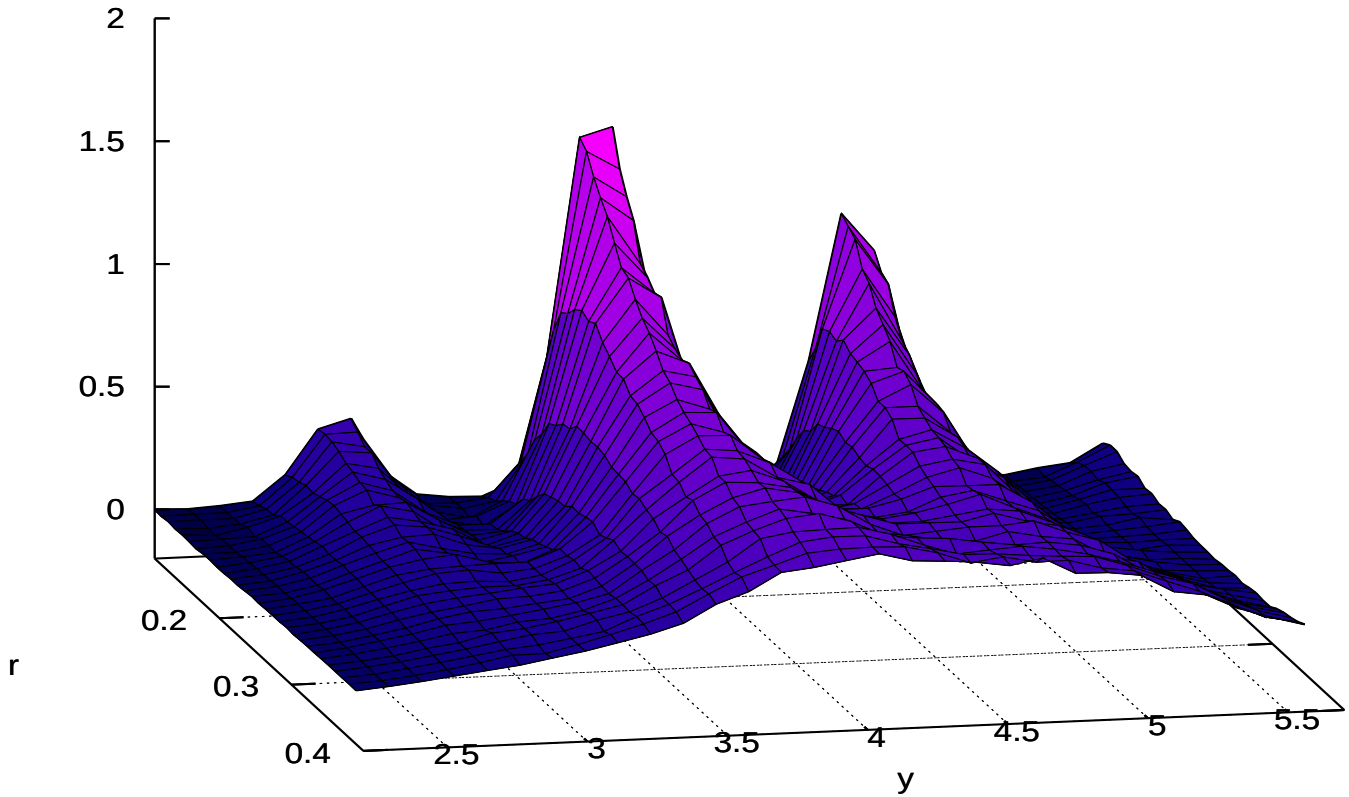
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Density estimation - localized Malliavin method

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Jump diffusion model [DJ], [DP04], [BMM]

- $(B_t)_{t\in\mathbb{R}_+}$ a standard Brownian motion,
- $(N_t)_{t\in\mathbb{R}_+}$ is a Poisson process with intensity $\lambda > 0$,
- $(Z_k)_{k\geq 1}$ an i.i.d. sequence of random variables with probability distribution $\nu(dx)$,
- $(X_t)_{t\in\mathbb{R}_+}$ a compound Poisson process with Lévy measure $\mu(dy) = \lambda\nu(dx)$. and finite intensity λ:

$$X_t = \sum_{k=1}^{N_t} Z_k, \qquad t \in \mathbb{R}_+, \tag{3}$$

- $(S_t^x)_{t\in\mathbb{R}_+}$ a jump-diffusion price process:

$$\begin{cases} \dfrac{dS_t^x}{S_t^x} = r(S_t^x)dt + \sigma_1(S_t^x)dB_t + \sigma_2(S_{t-}^x)dX_t, \\ \\ S_0^x = x. \end{cases}$$

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$$\frac{dS_t^\zeta}{S_t^\zeta} = rdt + \sigma_1 dB_t + \sigma_2(dN_t - \lambda dt), \qquad S_0^\zeta = x.$$

- $w \in L^2(\mathbb{R}_+)$ and

$$D_w f(B_{t_1}, \ldots, B_{t_n}, T_1, \ldots, T_n) = \sum_{i=1}^{n} \int_0^{t_i} w_s ds \frac{\partial f}{\partial x_i}(B_{t_1}, \ldots, B_{t_n}, T_1, \ldots, T_n).$$

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$$\frac{\partial}{\partial \sigma_2} \mathbf{E}\left[f(S_T^{\sigma_2})\right] = \mathbf{E}\left[f(S_T^{\sigma_2}) \frac{B_T}{\sigma_1 T}\left(\frac{N_T}{1+\sigma_2} - \lambda T\right)\right].$$

- Derivation with respect to absolutely continuous jump amplitudes [BMM].

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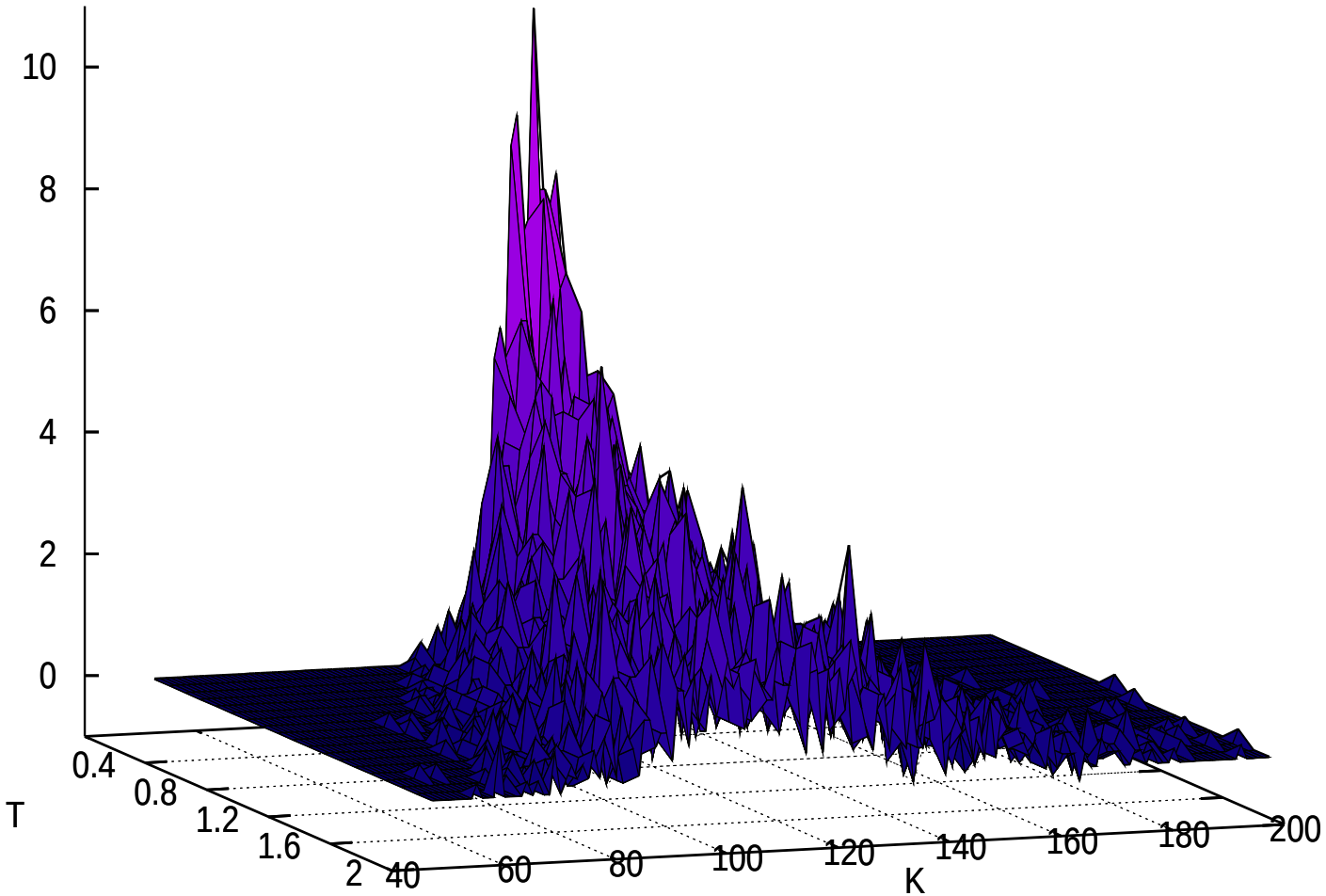
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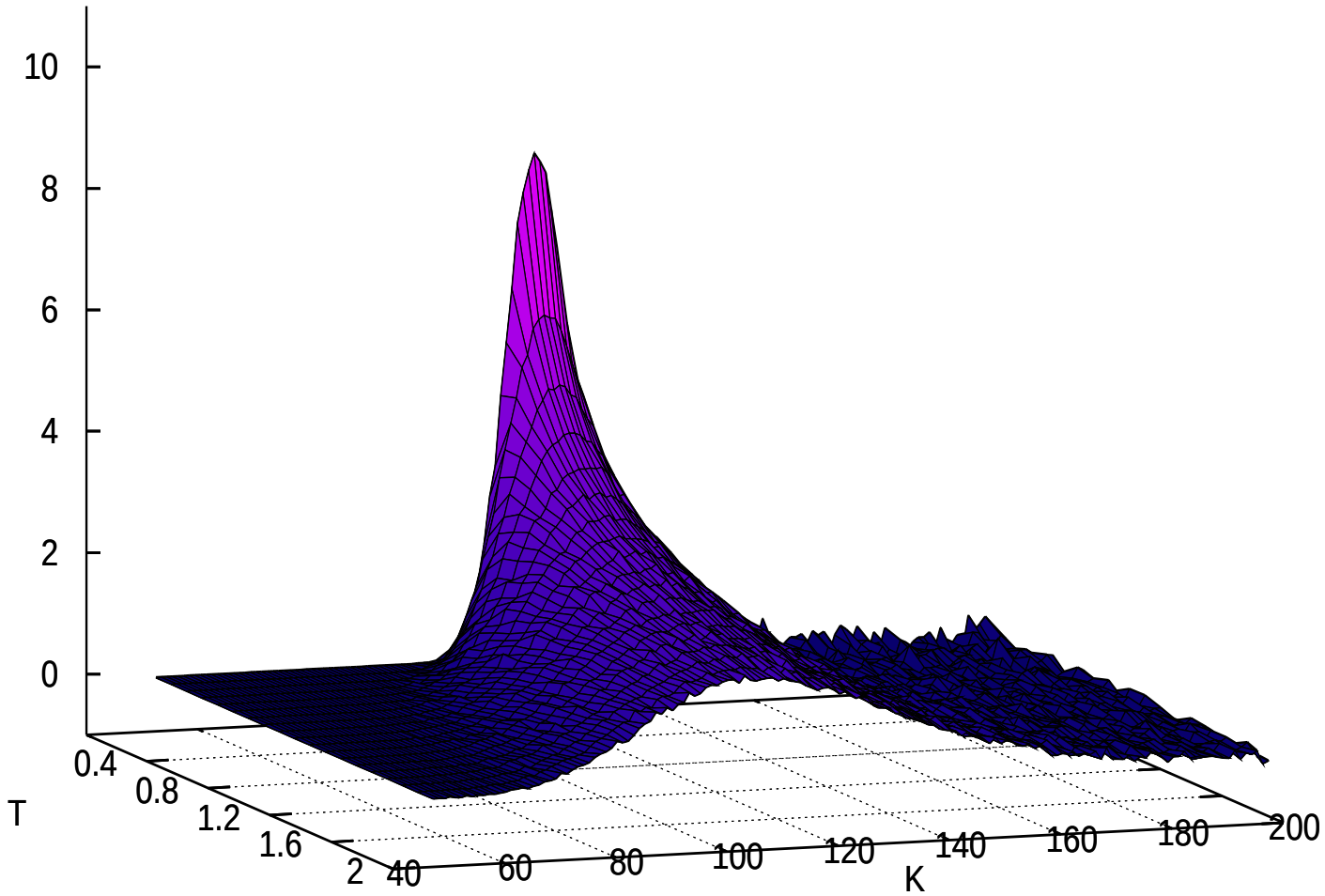
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Vega_2 as a function of K and T (finite differences, 10e4 samples, h=0.01)



Vega_2 as a function of K and T (Malliavin method, 10e4 samples, h=0.01)





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